

Generative Adversarial Networks for Cubism Painting Generation

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1 Hyperparameter Settings

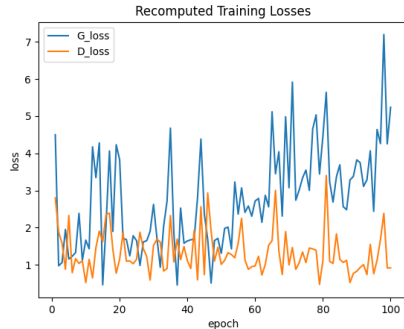
In this project, I initially implemented a standard Deep Convolutional GAN (DCGAN) to generate 128x128 Cubism paintings. However, the results was not satisfactory, so I subsequently adopted a Wasserstein GAN with Gradient Penalty (WGAN-GP). The detailed hyperparameter settings for both models are summarized and compared in Table 1.

Table 1: Hyperparameter Settings for DCGAN and WGAN-GP

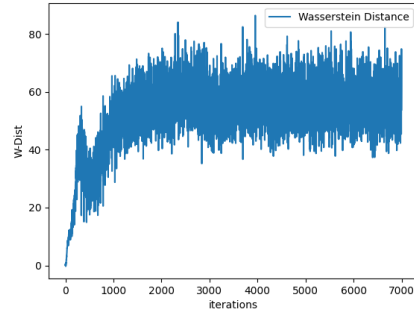
Hyperparameter	DCGAN (Baseline)	WGAN-GP (Proposed)
Optimizer	Adam	Adam
Learning Rate	0.0002	0.0001
Betas (β_1, β_2)	(0.5, 0.999)	(0.0, 0.9)
Batch Size	32	16
Latent Dimension (z)	128	128
Epochs	100	100
Critic Updates per G Update (n_{critic})	1	5
Gradient Penalty Weight (λ)	N/A	10.0
Loss Function	Binary Cross-Entropy (BCE)	Wasserstein Distance + GP

2 Training Loss Curves

Figure 1 illustrates the training dynamics of both models.



(a) DCGAN Loss Curve



(b) WGAN-GP Wasserstein Distance

Figure 1: Training progress of my DCGAN (left) and WGAN-GP (right).

As shown in Figure 1a, the standard DCGAN loss curve exhibits erratic fluctuations and extreme spikes, reflecting an unbalanced adversarial dynamic that directly causes mode collapse.

For the WGAN-GP model, I evaluate the training using the estimated Wasserstein Distance (W-Dist) (1) (Figure 1b). The curve initially spikes as the Critic rapidly distinguishes real paintings from noise, then plateaus into a stable oscillation, indicating the Generator reaches an equilibrium without mode collapse. However, the lack of a downward trend towards zero implies incomplete convergence; the model captures low-frequency color palettes, but learning high-frequency geometric lines requires more epochs or higher model capacity.

3 Inception Score and FID Score

The Inception Score (IS) and Fréchet Inception Distance (FID) of the two GANs are shown in Table 2. Clearly, the performance of WGAN-GP is better than DCGAN.

Table 2: Quantitative Evaluation of Generated Images (Epoch 100)

Model	FID Score ↓	Inception Score (IS) ↑
DCGAN	3.8564	1.1367
WGAN-GP	0.0900	1.7790

4 Examples of Generated Images

To evaluate the generation quality, I compare my generated 128x128 Cubism paintings from epoch 100 against real samples from the training dataset, as shown in Figure 2.

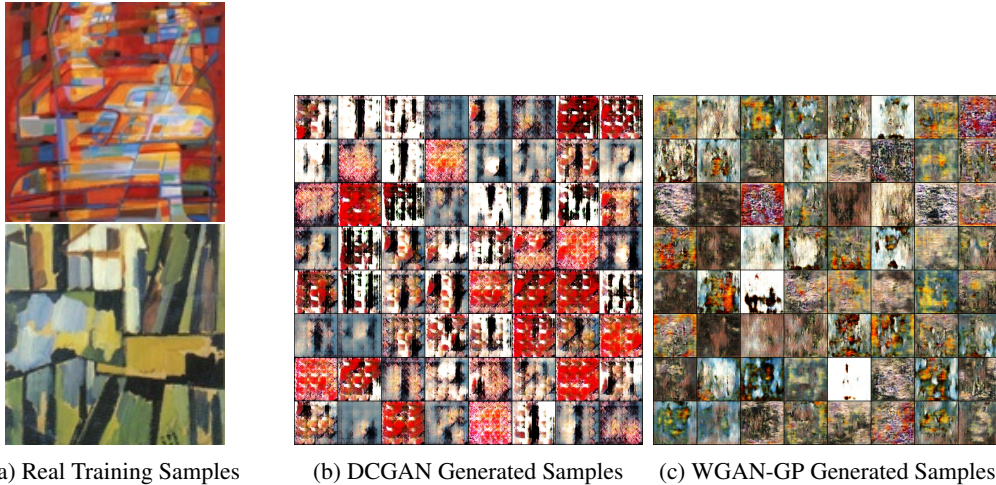


Figure 2: Visual comparison of real training data and my generated images after 100 epochs.

Compared to the real Cubism paintings, the standard DCGAN architecture fails to accurately capture the authentic color distributions. Furthermore, it suffers from severe mode collapse, repeatedly generating identical, meaningless patches without diversity.

In contrast, by implementing WGAN-GP, I successfully resolve the mode collapse, producing a diverse set of outputs with no repeating identical images. More importantly, the WGAN-GP effectively learns and replicates the rich color features and tonal variations present in the original training dataset. Nonetheless, due to low resolution of the training set and probably insufficient epochs, it fails to learn the geometric lines in Cubism paintings.

References

- [1] Gulrajani, I., Ahmed, F., Arjovsky, M., Dumoulin, V., & Courville, A. C. (2017). Improved training of wasserstein gans. *Advances in neural information processing systems*, 30.